

Regression Problem Overview

Regression means predicting a number.

MACHINE LEARNING SERIES



What Is Regression?

Regression is a type of Supervised Machine Learning — it predicts a numerical value.



House Price



Salary



Student Marks



Monthly Sales



Temperature

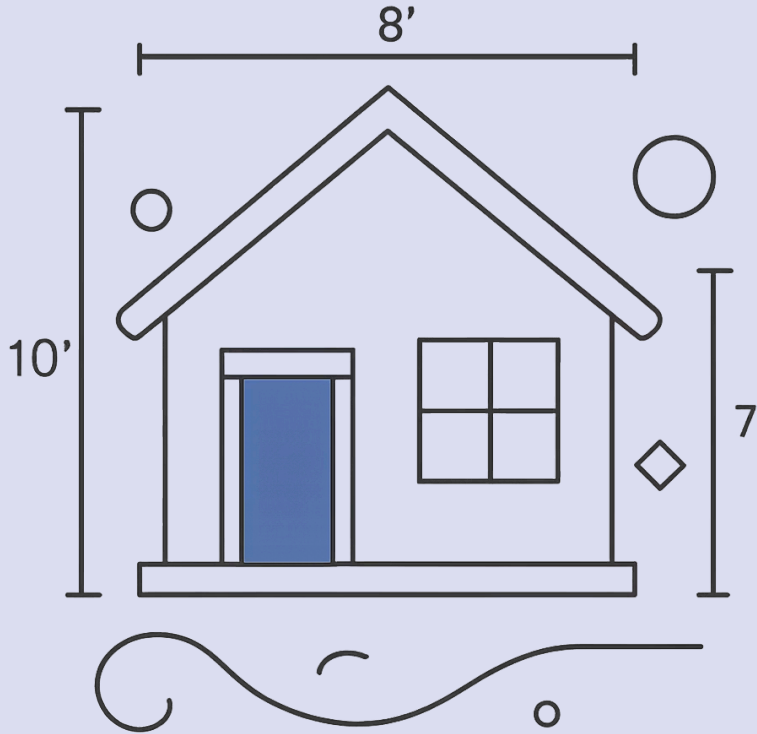


Delivery Time

Our Running Example: House Price Prediction

We use historical house data to predict the price of a new house.

Area	Bedrooms	House Age	Distance	Price
1000 sqft	2	5 yrs	8 km	₹55 lakh
1500 sqft	3	3 yrs	5 km	₹82 lakh
2000 sqft	4	2 yrs	4 km	₹120 lakh
900 sqft	2	10 yrs	12 km	₹42 lakh



Input and Output

Input Features

- Area
- Bedrooms
- House Age
- Distance from city

Target Variable

→ Price

i Area + Bedrooms + Age
+ Distance → Price

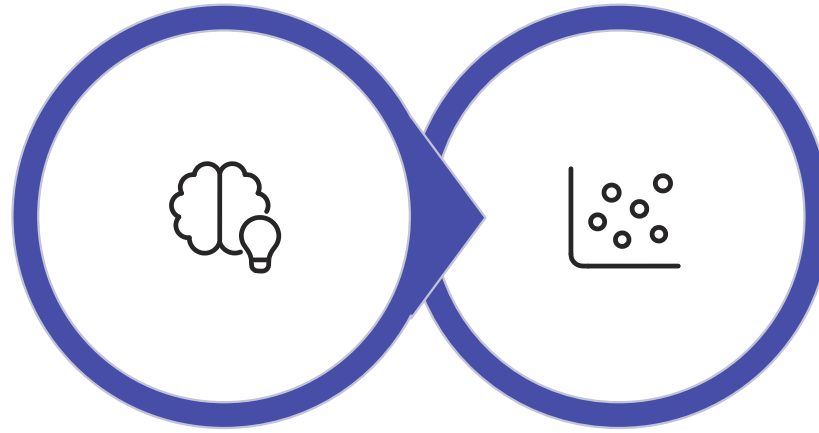
Why Is This Regression?

The output is a number:

Price = ₹90 lakh

- ✓ Simple Rule: If the output is a measurable number, it is usually regression.

Regression Belongs to Supervised Learning



Supervised
Learning

Regression

In our example: Input = Area, Bedrooms, Age, Distance → Answer = Price. Since the answer is available and numerical, this is Supervised Learning → Regression.

Easy Memory Line

How much?

House Price?

Salary?

Student Marks?

Next Month's Sales?

Delivery Time?

Regression vs Classification

Question	Output	Type
What is the house price?	₹90 lakh	Regression
Will the loan be approved?	Yes / No	Classification
What are the final marks?	82	Regression
Will the student pass?	Pass / Fail	Classification
What is tomorrow's temperature?	32°C	Regression
Is this email spam?	Spam / Not Spam	Classification

Regression = Number

Classification = Category

Student Example — Regression

Goal: Predict final marks — output is a number, so this is Regression.

Study Hours	Attendance	Previous Marks	Final Marks
2	60%	40	45
5	85%	70	78
6	90%	80	88

 Target = Final Marks → a number → Regression

Student Example — Classification

Goal: Predict pass or fail — output is a category, so this is Classification.

Study Hours	Attendance	Previous Marks	Result
2	60%	40	Fail
5	85%	70	Pass
6	90%	80	Pass



Target = Result → a category → Classification

Regression Is an Estimate

Regression predictions are usually not exact — but closeness matters.

✓ Good Prediction

Actual: ₹92 lakh

Predicted: ₹90 lakh

Close enough — model is useful.

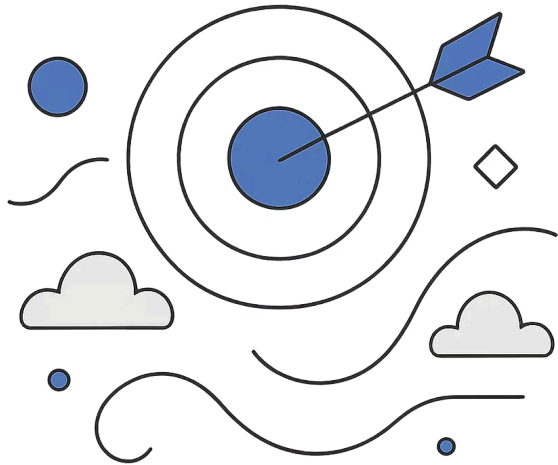
✗ Poor Prediction

Actual: ₹92 lakh

Predicted: ₹60 lakh

Error is large — model needs improvement.

What Is Error?



Error = Actual – Predicted

- ❏ Actual price = ₹100 lakh
Predicted price = ₹92 lakh
Error = ₹8 lakh

Regression models are evaluated by checking: How far are predictions from actual values?

Small Error vs Large Error

Actual Price	Predicted Price	Error
₹100 lakh	₹98 lakh	₹2 lakh ✓
₹100 lakh	₹92 lakh	₹8 lakh ⚠
₹100 lakh	₹60 lakh	₹40 lakh ✗

We measure error using: MAE, MSE, RMSE, R^2 Score — covered in later topics.

Real-Life Example: Salary Prediction

Input Features

- Experience
- Education
- Skills score
- Interview score
- Location

Target

Salary — a number →
Regression



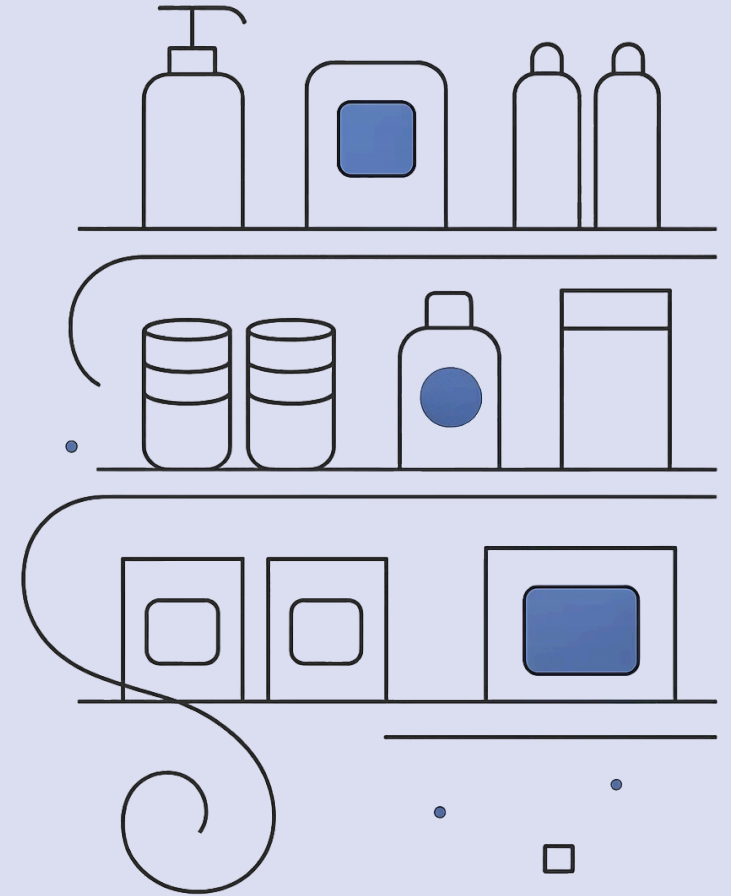
Real-Life Example: Sales Prediction

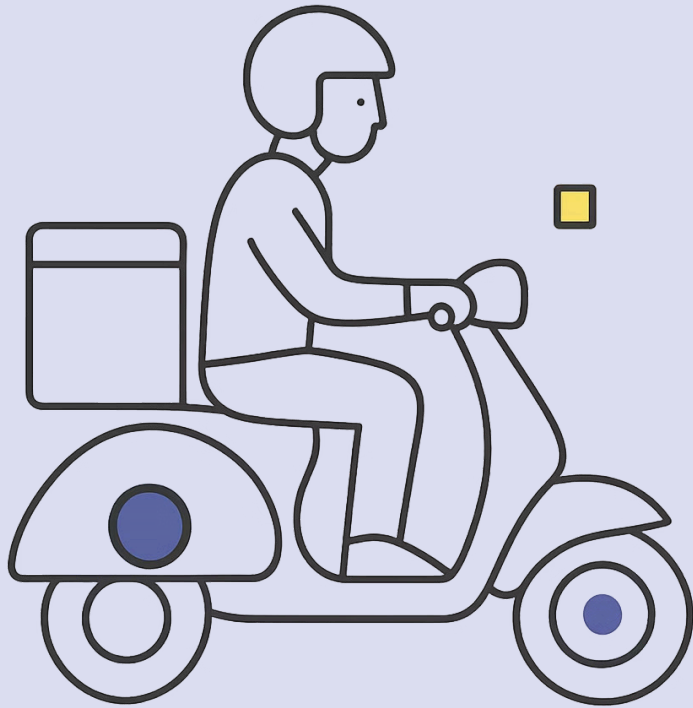
Input Features

- Month & product category
- Festival season
- Discount & ad spend
- Previous sales

Target

Sales amount — numerical →
Regression





Real-Life Example: Delivery Time Prediction

Input Features

- Distance & traffic level
- Weather & order size
- Delivery partner availability

Target

Delivery time in minutes — a number → Regression

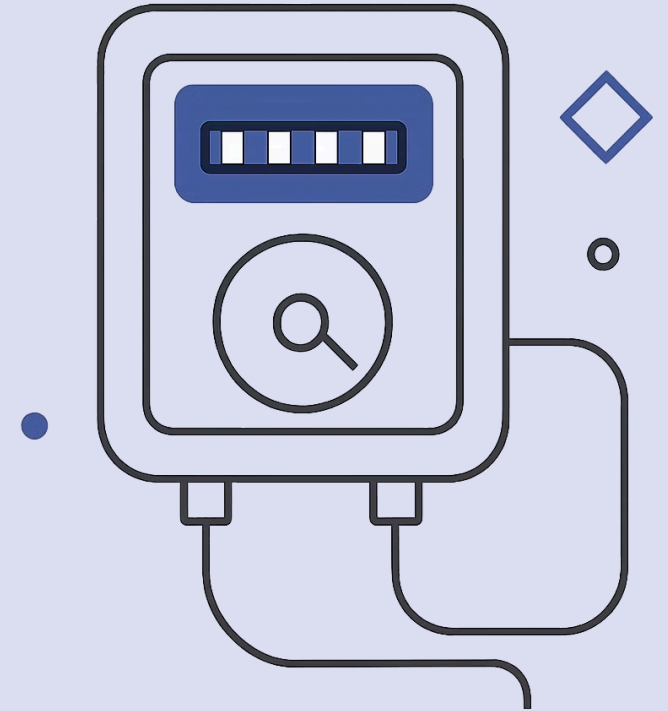
Real-Life Example: Electricity Consumption

Input Features

- Temperature & day of week
- Number of people
- Appliance usage
- Previous consumption

Target

Electricity units consumed —
numerical → Regression



Regression Mental Model



Think Like a Real Estate Agent

They look at area, location, age, facilities, and market trends — then estimate:

"This house may sell for around ₹90 lakh."

That is regression thinking.

Regression Output Can Be Decimal

The output can be an integer or a decimal — as long as it is a measurable numerical value.

32.7°C

Temperature

42.5

Delivery (mins)

₹87.6L

House Price

₹12.35L

Sales Forecast

Beginner Confusion

If the output is a number, is it always regression?

Mostly yes — but be careful. Some numbers are actually labels.

 Looks like a number...

- Class 1, Class 2, Class 3
- Pincode
- Product ID
- Customer ID

But they are categories or identifiers

These numbers represent labels, not measurable quantities — so they are not regression targets.

The Better Rule

Measurable quantity → Regression		Label / Category → Classification
Value	Meaning	ML Type
Price	Measurable quantity	Regression
Temperature	Measurable quantity	Regression
Class 1/2/3	Category	Classification
Product ID	Identifier	Not a useful target

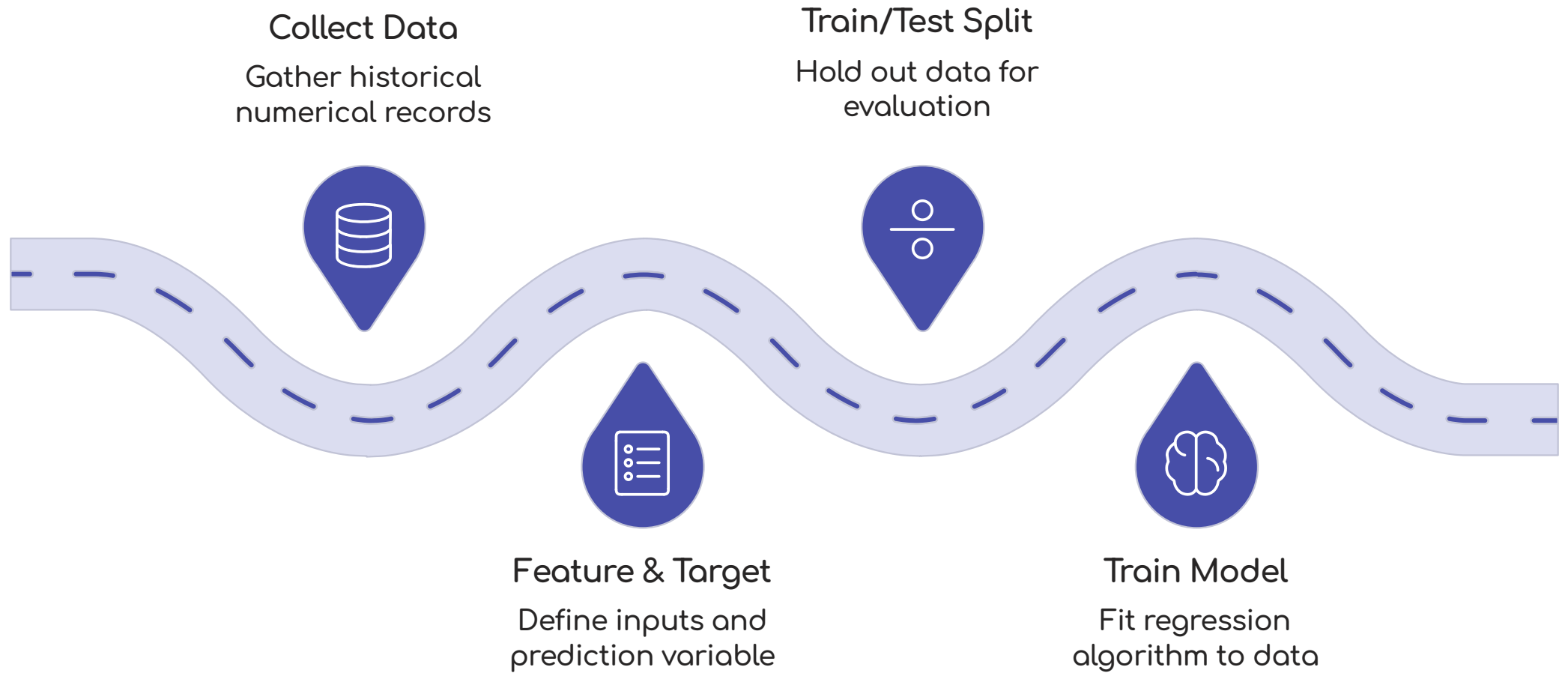
One Feature or Many Features?

 Simple Linear Regression
One feature:
Area → Price

 Multiple Linear Regression
Many features:
Area + Bedrooms + Age + Distance → Price

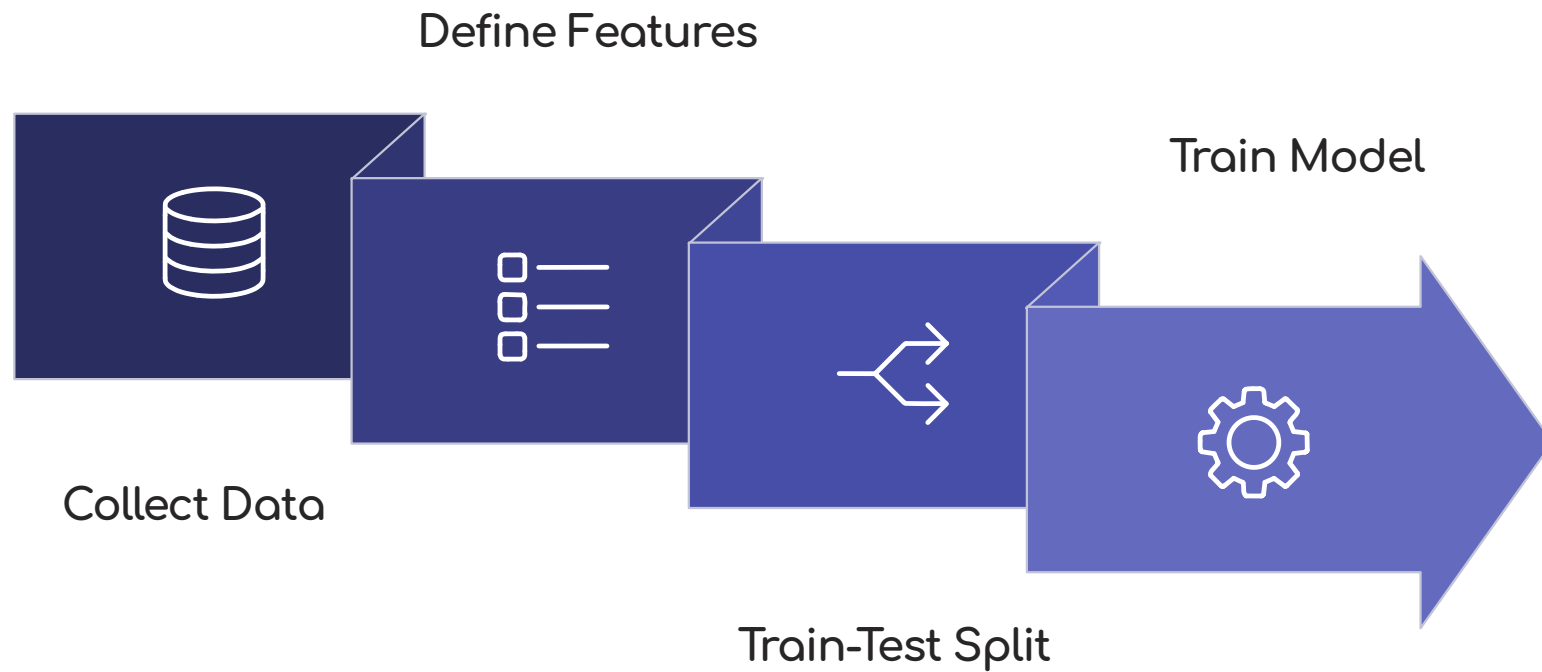
Regression works with one or many input features — the key is that the output is always a number.

Regression Project Flow



Every regression project follows this structured pipeline from raw data to a refined, accurate model.

House Price Regression Flow



Applying the regression pipeline to our running house price example.

Python Preview

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X = df[["area", "bedrooms", "age", "distance"]]
y = df["price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
prediction = model.predict(X_test)
```

✔ We trained a regression model to predict house price from four input features.

Final Summary

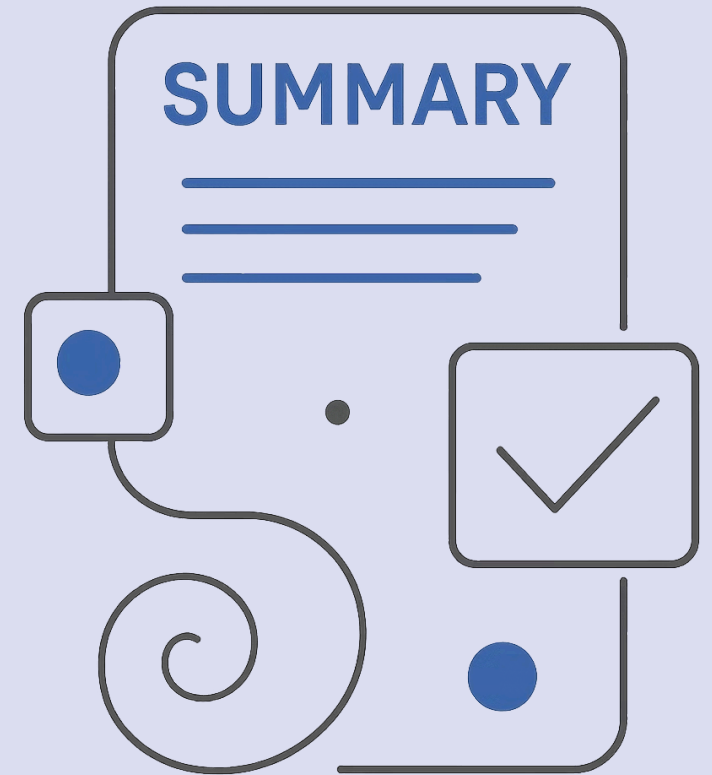
Regression = How much?

House price, salary, marks, sales, temperature, delivery time, electricity usage

Classification = Which category?

Pass/fail, spam/not spam, approved/rejected

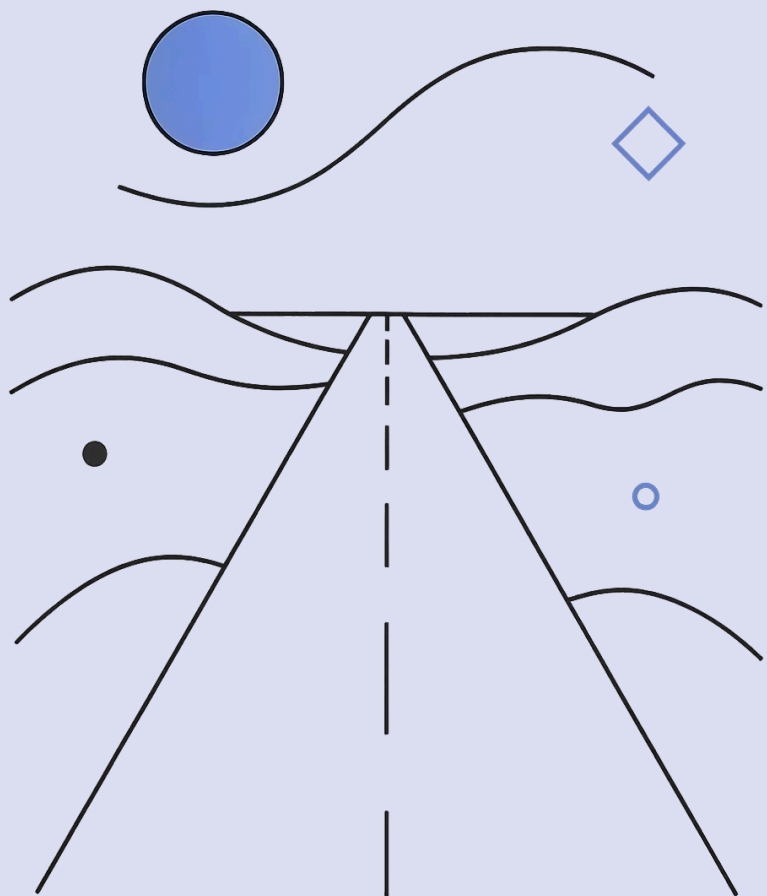
 Core rule: If the output is a measurable number → Regression.



Quick Quiz

Identify whether each problem is Regression or Classification:

#	Problem	Answer
1	Predict house price	Regression
2	Predict pass or fail	Classification
3	Predict monthly sales	Regression
4	Predict spam or not spam	Classification
5	Predict delivery time	Regression



Up Next: Simple Linear Regression

Now that we understand regression, the next topic is:

Simple Linear Regression

We will learn how a model predicts a number using one input feature.



Example: Area $\rightarrow$ House Price